

DEMAND FORECASTING

Baseline Approach & Documentation

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Approach: AutoARIMA Time Series Forecasting

Forecast Horizon: 90 Days

1. Executive Summary

This document outlines the baseline approach used for demand forecasting across a multi-store, multi-item retail environment. The solution utilizes **AutoARIMA**, a statistical forecasting model, to predict sales for 90 days into the future. The model was trained on 5 years of historical sales data at the item-store level and achieved reasonable performance metrics with RMSE of ~14 and MAPE of ~24%.

Metric	Value
Dataset Size	5 years of daily sales data
Number of Stores	50
Number of Items	Multiple items per store
Forecast Horizon	90 days
Primary Model	AutoARIMA
RMSE	~14
MAPE	~24%

2. Project Overview

Objective: Forecast daily sales for each item-store combination 90 days into the future to support inventory planning, resource allocation, and business decision-making.

Problem Type: Multi-variate time series forecasting with hierarchical data structure (item-store level).

Approach: Statistical forecasting using AutoARIMA, which automatically selects optimal ARIMA(p,d,q) parameters for each series based on information criteria (AIC/BIC).

3. Data Description

3.1 Dataset Characteristics

Training Data: 5 years of historical daily sales data

Time Period: Daily observations from multiple years

Granularity: Item-Store level (combination of SKU and location)

Geographic Scope: 50 retail stores

Key Variables: Date, Store ID, Item ID, and Sales volume

3.2 Data Quality

Missing Values: Minimal missing values in the dataset; no significant gaps in the time series.

Data Completeness: Complete daily records across all store-item combinations.

Outliers: The data exhibits normal seasonal patterns with consistent variance across years.

Stationarity: Time series analysis revealed **stationary** behavior with consistent mean, variance, and reproducible seasonality patterns.

4. Exploratory Data Analysis

Temporal Patterns: Analysis of sales trends across different item-store combinations revealed strong yearly seasonal patterns. The data exhibits recurring cycles, indicating that products follow annual seasonality rather than daily or weekly patterns.

Sales Behavior:

- **Seasonal Component:** Strong yearly seasonality (365-day cycles) observed across products
- **Trend:** Series are primarily stationary with minimal long-term trend
- **Variance:** Relatively constant variance throughout the observation period
- **Mean Reversion:** Sales revert to consistent mean levels, supporting ARIMA assumptions

Verdict: The stationary nature and consistent seasonality patterns make this data suitable for ARIMA-based forecasting methods.

5. Data Preprocessing & Feature Engineering

5.1 Data Preparation

Encoding: Each item-store combination was encoded into a unique identifier for model training. This hierarchical ID structure enables the model to learn patterns specific to each product-location pair.

Column Renaming: Standard column names were applied for compatibility with the statsforecast library:

- *date* → *ds* (datestamp)
- *sales* → *y* (target variable)
- *item_store* → *unique_id* (series identifier)

5.2 Feature Engineering

Temporal Features Extracted:

- **Year:** Calendar year extracted from date
- **Month:** Month of the year (1-12)
- **Day of Week:** Numeric representation (0=Monday, 6=Sunday)

Lag Features:

- **Lag-1:** Sales from 1 day prior (captures immediate temporal dependence)
- **Lag-7:** Sales from 7 days prior (captures weekly patterns)

Missing Value Handling: Lag features introduced NaN values for the first observations; these were filled with 0 to maintain data integrity.

Feature Purpose: These features serve as confounding variables in the context of AutoARIMA but are primarily used for the ML baseline model. For pure statistical forecasting, AutoARIMA relies on the endogenous series (y) itself.

6. Model Selection & Training

6.1 Model Architecture

Primary Model: AutoARIMA

AutoARIMA is a statistical forecasting method that automatically selects optimal ARIMA(p , d , q) parameters for each time series by:

- Testing multiple combinations of parameters
- Using information criteria (AIC/BIC) for model selection
- Applying differencing to achieve stationarity if needed

Baseline Model: Seasonal Naive

A seasonal naive model was also included as a baseline for comparison. This model simply repeats the value from one year ago (365-day season length).

6.2 Training Strategy

Implementation: The StatsForecast library was used for efficient model training across all 250 series (50 stores $\times$ 5 items) simultaneously.

Cross-Validation Approach:

- **Type:** Time series cross-validation (walk-forward)
- **Horizon (h):** 90 days (matches the production forecast horizon)
- **Step Size:** 7 days (weekly steps for validation windows)
- **Number of Windows:** 3 validation windows
- **Parallel Processing:** `n_jobs=-1` for multi-core execution

Rationale: Cross-validation ensures the model generalizes well to unseen data and prevents overfitting while respecting temporal ordering.

7. Performance Metrics & Evaluation

7.1 Metric Definitions

Three key metrics were computed to evaluate forecast accuracy:

1. RMSE (Root Mean Squared Error)

Measures the average magnitude of prediction errors. Penalizes larger errors more heavily.

Interpretation: A lower RMSE indicates better accuracy. RMSE is in the same units as the target variable (sales).

2. MAPE (Mean Absolute Percentage Error)

Expresses prediction error as a percentage of actual values. Scale-invariant and interpretable across different series.

Interpretation: Lower MAPE% indicates better accuracy. Typical baseline is 20-30% for retail demand.

3. MAE (Mean Absolute Error)

Average absolute difference between predicted and actual values. More robust to outliers than RMSE.

Interpretation: Lower MAE indicates smaller average prediction errors.

7.2 Cross-Validation Results

AutoARIMA Performance:

- RMSE: ~14 units
- MAPE: ~24%
- MAE: Calculated during cross-validation

Interpretation: The AutoARIMA model achieves reasonable baseline performance with an average prediction error of approximately 24% relative to actual sales values. This is acceptable for a baseline approach but leaves room for improvement through:

- Ensemble methods combining statistical and ML models
- Advanced feature engineering capturing external factors
- Incorporating exogenous variables (promotions, holidays, etc.)

8. Forecasting Results

Test Set Predictions: After training on the complete historical dataset, the AutoARIMA model was applied to generate 90-day ahead forecasts for each item-store combination.

Output: A submission file containing:

- Item ID
- Store ID
- Date (days 1-90 into the future)
- Predicted Sales (AutoARIMA forecast)

Coverage: Complete forecasts for all 250 item-store combinations across the entire 90-day horizon.

9. Key Findings & Insights

✓ **Data Quality:** Stationary time series with consistent yearly seasonality make the data well-suited for ARIMA-based forecasting.

✓ **Model Suitability:** AutoARIMA effectively captures the ARIMA structure of individual item-store series, automatically selecting appropriate (p,d,q) parameters.

✓ **Baseline Performance:** The 24% MAPE represents a solid baseline for retail demand forecasting, though there is clear room for enhancement.

✓ **Scalability:** The statistical approach handles 250 series efficiently and can scale to larger product/location combinations.

10. Future Improvements & Next Steps

1. Ensemble Methods

Combine AutoARIMA with machine learning models (Random Forest, XGBoost) to leverage both statistical and learned patterns.

2. Exogenous Variables

Incorporate external features such as:

- Promotional calendars
- Holiday and special event flags
- Weather data (if location-specific)
- Store-level characteristics

3. Advanced Feature Engineering

- Rolling statistics (moving averages, volatility)
- Fourier features for capturing seasonality
- Item-store interaction terms
- Trend decomposition

4. Hierarchical Forecasting

Reconcile forecasts at multiple aggregation levels (by item, by store, by category) to ensure coherent predictions across hierarchies.

5. Model Tuning & Hyperparameter Optimization

- Experiment with different season lengths for seasonal patterns
- Tune cross-validation parameters

- Test multiple baseline models systematically

6. Deep Learning Approaches

Explore LSTMs, Transformers, and other neural architectures for capturing complex temporal dependencies.